

11. Recommendations

Based on findings from exploratory data analysis, correlation analysis, machine learning modelling, SHAP feature importance analysis, and dashboard insights, recommendations are developed for two key stakeholders: potential Airbnb investors and existing Airbnb hosts.

The analysis demonstrates that Airbnb performance is primarily influenced by location, property characteristics, and customer demand patterns, while host-related factors and amenities provide additional opportunities for improving competitiveness.

For Investors

✓ **Prioritise High-Demand Locations**

The analysis identified location as the strongest determinant of Airbnb pricing performance. Both correlation analysis and SHAP feature importance demonstrated that geographical features, particularly latitude, were the most influential predictors of listing prices. This indicates that investment decisions should focus heavily on where the property is located, as location determines customer demand, pricing potential, and long-term revenue opportunities. Investors should prioritise properties located in high-demand cities, popular tourist destinations, central neighbourhoods, areas with strong review activity, locations close to attractions, transportation, and business centres. Neighbourhood-level analysis should be conducted before investment decisions because significant performance differences exist within the same city.

✓ **Select Properties with Strong Revenue Potential**

The machine learning model identified bedrooms and accommodation as major contributors to predicted listing prices. Larger properties generally achieve higher prices because they attract larger customer groups, including families and group travellers. However, higher purchase costs do not always guarantee better returns. Before making decisions, investors should evaluate properties based on the expected revenue potential, customer demand, property capacity, competitive pricing, and operating costs. A moderately sized property in a high-demand neighbourhood may generate stronger returns than a premium property in a low-demand location.

✓ **Consider Demand Stability and Seasonality**

Dashboards revealed that customer demand varies significantly between cities and neighbourhoods. Some locations maintain consistent demand, while others experience stronger seasonal fluctuations. So, investors should evaluate historical demand trends, seasonal patterns, review growth, and market competition. The locations with stable year-round demand may provide more predictable returns compared with highly seasonal markets.

✓ **Use Predictive Analytics Before Investment Decisions**

The Random Forest model achieved the strongest predictive performance among tested models, explaining approximately 75% of price variation. Before purchasing an Airbnb property, investors can use predictive models as a reference to estimate expected listing prices, compare similar properties, identify undervalued opportunities, and evaluate potential revenue outcomes. This allows investors to make decisions based on market evidence rather than relying only on intuition or historical assumptions.

For Hosts

✓ **Optimise Pricing Based on Market Position**

The analysis shows that Airbnb prices are strongly influenced by location, property size, and market demand. Hosts should regularly review competitor listings and adjust pricing strategies accordingly. Hosts can implement dynamic pricing strategies by increasing prices during peak demand periods, offering competitive rates during low seasons, comparing prices with similar properties nearby, and adjusting prices based on occupancy trends. This can help maximise revenue while maintaining competitiveness.

✓ **Improve Property Features That Influence Customer Value**

The model identified property characteristics such as bedrooms and accommodation capacity as important pricing factors. While hosts cannot always change property size, they can improve how the property is presented and utilised. Hosts can consider improving room functionality, creating attractive spaces for groups and families, highlighting accommodation capacity clearly, and providing suitable facilities for target customer segments. For example, a property designed for families or groups can justify higher pricing compared with similar-sized properties targeting individual travellers.

✓ **Strengthen Customer Experience and Reviews**

Although host-related features had a smaller impact on price prediction, customer satisfaction remains important for maintaining competitiveness. Positive reviews influence customer trust, listing visibility, and future demand. Guest experience can be improved through faster communication, accurate listing descriptions, high-quality photos, clean and well-maintained properties, essential amenities, and smooth check-in processes. These improvements can increase review scores and improve long-term booking performance.